Research an Presenting

06 – Latex Tutorial

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Overview of today

- What is LaTeX?
- Example 1: Basic document
- Where to use LaTeX/Setup
- Basics of LaTeX

What is latex?

- LaTeX (pronounced “LAY-tek” or “LAH-tek”)
- A typesetting language, think a programming language to make documents beautiful
- It is built for anything mathematical related (research papers especially)
- Microsoft word is built using “ WYSWYG - what you see is what you get” mantra of formatted text
- LaTeX is written in plain text with markup tagging
- Why use LaTeX? Have you ever had the feeling when using word or docs that you just couldn’t get something to be in the exact right place? With LaTeX you can.

Example 1: Basic document

```latex
\documentclass{article} % Starts an article
\usepackage{amsmath} % Imports amsmath
\title{\LaTeX} % Title

\begin{document} % Begins a document
  \maketitle
  \LaTeX{} is a document preparation system for
  the \TeX{} typesetting program. It offers
  programmable desktop publishing features and
  extensive facilities for automating most
  aspects of typesetting and desktop publishing,
  including numbering and  cross-referencing,
  tables and figures, page layout,
  bibliographies, and much more. \LaTeX{} was
  originally written in 1984 by Leslie Lamport
  and has become the  dominant method for using
  \TeX; few people write in plain \TeX{} anymore.
  The current version is \LaTeXe.

  % This is a comment, not shown in final output.
  % The following shows typesetting  power of LaTeX:
  \begin{align}
    E_0 &= mc^2 \\
    E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
  \end{align}
\end{document}
```

LaTeX

LaTeX is a document preparation system for the TeX typesetting program. It offers programmable desktop publishing features and extensive facilities for automating most aspects of typesetting and desktop publishing, including numbering and cross-referencing, tables and figures, page layout, bibliographies, and much more. LaTeX was originally written in 1984 by Leslie Lamport and has become the dominant method for using TeX; few people write in plain TeX anymore. The current version is LaTeX 2_ε.

$$E_0 = mc^2 \tag{1}$$

$$E = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} \tag{2}$$

How to setup Latex?

Online	LaTeX online services like Overleaf, Papeeria, CoCalc or Prism
Linux	TeX Live distribution
Mac OS	MacTeX distribution
Windows	MiKTeX or TeX Live distributions

LaTeX editor TeXStudio

How Latex Work

User Input
- main.tex
- references.bib
- images (.png, .jpg, .pdf and .eps)

Auxiliary File
- .aux (Auxiliary file)
- .log (Log file)
- .toc (Table of contents data)
- .lof (List of figures data)
- .lot (List of tables data)
- .out (General output info)
- .idx (Raw index data)

Latex Engine
pdflatex / xelatex / lualatex

Document Structure
- Document Class (.cls)
- Packages (.sty)

Output
- main.pdf
- main.dvi

Basics of latex: important chars

- There are some important characters and a characteristic of LaTeX
- Characters:
  - “\”: Starts a typesetting command
  - “{}”: Any arguments are put inside the curly braces
  - ‘\\”: Newline without a new paragraph
  - “%”: Used to start a comment, everything after on the same line is treated as such
- Characteristic:
  - LaTeX wraps adjacent lines as if they were part of a same paragraph. Insert extra return for a new paragraph

Source:
```
This is one paragraph.

This is another.
```

Output:

This is one paragraph. This is another.

Basic Document Structure

- The format of a document is pretty simple.
- In the preamble
  - document class
  - packages
- In the front matter
  - Title/author
  - Table of contents/ List of figures
- In the body
  - Contents
- In the back matter
  - bibliography

Basics of latex: document class

- Document classes is how LaTeX interprets a particular document
- Main available types are:
  - **Article:** articles in scientific journals, short reports, program documentation
  - **Report:** longer reports containing several chapters, small books, thesis work
  - **Slides/beamer:** for presentations
- Starting a new document:

```
\documentclass [options] {style}

\begin{document}
%your writing here
\end{document}
```

Predefined classes like article, report, letter or book

Options within a class such as Paper size, font size etc

Basics of latex: Packages

- Just like any other programming language, people can build packages that give the programming language to do certain things and other people can download said packages.
- Most packages are already installed by default
- To use them, do a `\usepackage[options]{packagename}` before you begin your document

Basics of latex: Sectioning

- Sectioning is to create different sections and subsections in your documents.
  - **\section{name of section}**: creates a section with the specific name, is numbered
  - **\subsection{name of subsection}**: creates a subsection under the larger section
  - **\subsubsection{name of subsubsection}**: I think you get the idea
  - **\section*{Name of section}** creates a section with the specific name, without the number.
  - **\paragraph{name of paragraph}**: like section but not numbered
  - **\subparagraph{name of subparagraph}**: like paragraph but indented, goes under a paragraph

Basics of latex: Sectioning

- \section{name of section}

  2 Name of section

  creates a section with the specific name, is numbered

- \subsection{name of subsection}

  2.1 Name of subsection

  creates a subsection under the larger section

- \subsubsection{name of subsubsection}

  2.1.1 Name of subsubsection

  I think you get the idea

- \paragraph{name of paragraph}

  Name of paragraph like section but not numbered

- \subparagraph{name of subparagraph}

  Name of subparagraph like paragraph but indented, goes under a paragraph

- \section*{Name of section}

  Name of section

  creates a section with the specific name, without the number.

Basics of Latex: Tables

- Simple tables can be produced by **\begin{tabular}[pos]{tablespec}**
- Within the **{tablespec}** section, one details the number of columns, the alignment, and the number of vertical lines of the table. **{lrc}, {|l|r|c}**
- Then type in from left to right, the values for each cell with & in between.
- Put "\\" at the end of each row, then input another row of values if needed.
- \hline

```
\section {Table}
    Here is table~\ref{tab:table1} in the text.

\begin{table}[h!]
    \begin{center}
        \caption{Your first table.}
        \label{tab:table1}
        \begin{tabular}{l|c|r} % <-- Alignments: 1st column left, 2nd middle and 3rd right, with vertical lines in between
            \textbf{Value 1} & \textbf{Value 2} & \textbf{Value 3}\\
            $\alpha$ & $\beta$ & $\gamma$ \\
            \hline
            1 & 1110.1 & a\\
            2 & 10.1 & b\\
            3 & 23.113231 & c\\
        \end{tabular}
    \end{center}
\end{table}
```

Here is table 1 in the text.

Table 1: Your first table.

Value 1	**Value 2**	**Value 3**
α	β	γ
1	1110.1	a
2	10.1	b
3	23.113231	c

Basics of Latex: Image

- Using the *figure environment* and the *graphicx package*
- `\usepackage{graphics}`

h (here) – same location
t (top) – top of page
b (bottom) – bottom of page
p (page) – on an extra page
! (override) – will force the specified location

```
\begin{figure}[t]
    \centering
    \includegraphics[width=0.5\linewidth]{cat.jpg}
    \caption{A Cute Cat.}
    \label{fig:boat1}
\end{figure}
```

Figure 1: A Cute Cat.

This is a sample text. Figure 1 shows a cute cat.

Basics of Latex: Equation

- allows two writing modes for mathematical expressions: the ***inline*** math mode and ***display*** math mode

```
\section{Equation}

An inline equation: $a^2 + b^2 = c^2$.

\begin {equation}
    g_{ \gamma_{n}}(t) =
    \frac{1}{
    \sqrt{s_{n}}}\;g(\frac{t-p_{n}}{s_{n}})\,\exp\left\{j(2\pi
    f_{n}t+\phi_{n})\right\},
    \label {eq1}
\end {equation}


\begin{align}
    E_0 &= mc^2 \\
    E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
\end{align}
```

An inline equation: $a^2 + b^2 = c^2$.

$$g_{\gamma_n}(t) = \frac{1}{\sqrt{s_n}}\;g(\frac{t-p_n}{s_n})\,\exp\left\{j(2\pi f_n t+\phi_n)\right\}, \tag{1}$$

$$E_0 = mc^2 \tag{2}$$

$$E = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} \tag{3}$$

Basics of Latex: lists

- Can create ordered or unordered lists or sublists as shown on the right.

```
Unordered list:
\begin{itemize}
    \item First item
    \item Second item
\end{itemize}

Ordered list:
\begin{enumerate}
    \item First item
    \item Second item
    \item Third item
\end{enumerate}
```

Unordered list:

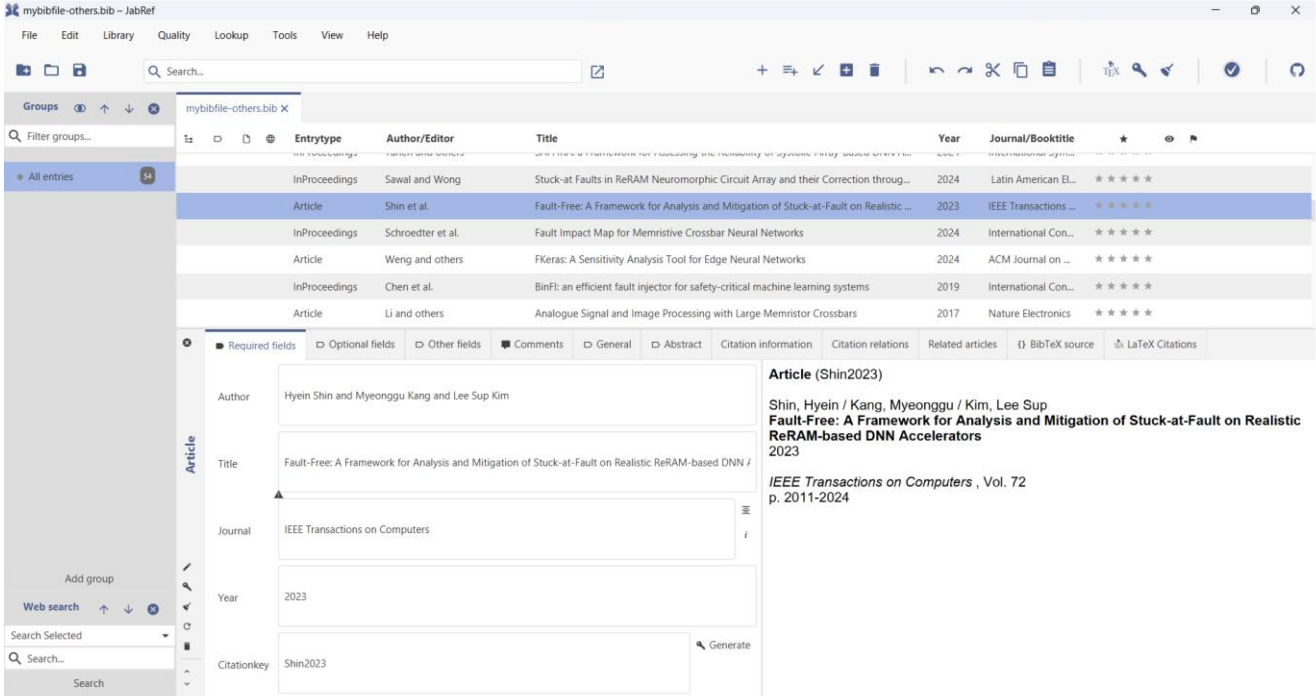
- First item
- Second item

Ordered list:

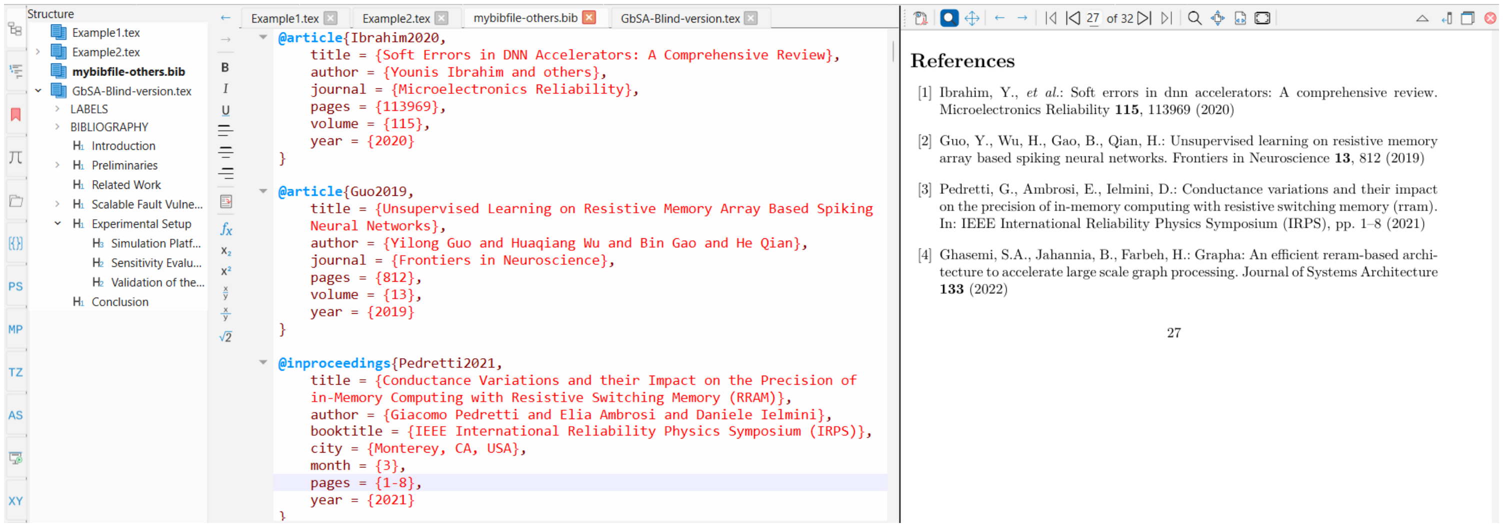
1. First item
2. Second item
3. Third item

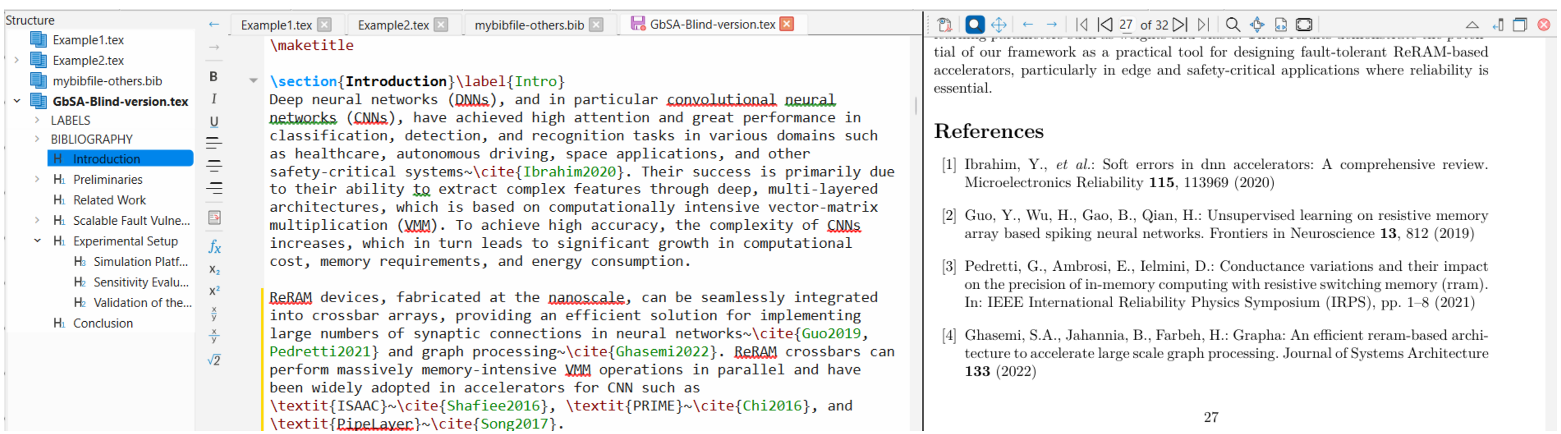
Basics of Latex: Citations

- \cite{bibtexkey}, citeyear{bibtexkey}
- •It is more convenient to create a bibliography file, called BibTeXfile(.bib), and use it as needed.
- JabRef (http://jabref.sourceforge.net/)



Basics of Latex: Citations






```
\bibliography{mybibfile-others} % common bib file

\end{document}
```

Advanced Latex: circuit diagrams/syntax highlighting

- Include this because most of us here are EE/Compe majors
- Can use the **circuitikz** package to create circuit diagrams
- Can use the listings package to use syntax highlighting for various languages that you include code snippets in your document.

```
\begin{circuitikz}
  \draw (0,0)
  to[V,v=$U_q$] (0,2) % The voltage source
  to[short] (2,2)
  to[R=$R_1$] (2,0) % The resistor
  to[short] (0,0);
  \draw (2,2)
  to[short] (4,2)
  to[L=$L_1$] (4,0)
  to[short] (2,0);
  \draw (4,2)
  to[short] (6,2)
  to[C=$C_1$] (6,0)
  to[short] (4,0);
\end{circuitikz}
```

This would give us the following diagram:

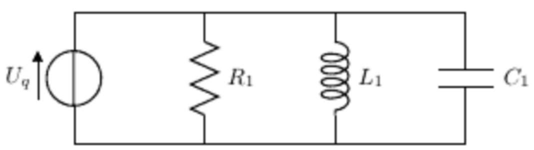


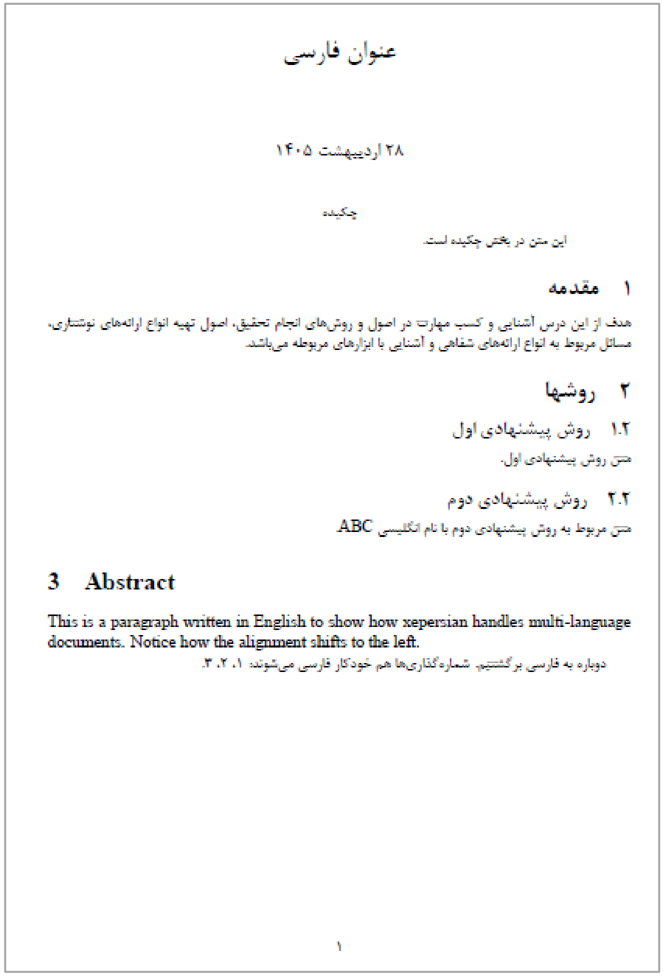
Figure 1: My first circuit.

Persian Packages in LaTeX

- **xepersian** (Recommended and Standard Package)
  - Full Persian support: Automatically handles right-to-left text, Persian numerals, Persian footnotes, and the Persian calendar
  - System fonts: Allows the use of system fonts such as B Nazanin, etc
  - Requires **XeLaTeX**: Documents must be compiled with the XeLaTeX engine
- **Babel** (Modern and International Approach)
  - International standard: Works well with modern engines such as LuaLaTeX
  - Good for multilingual projects: Suitable when documents contain multiple languages.
  - Less optimized for Persian users: Persian configuration is not as ready-to-use as xepersian.
- **ArabTeX (Old and Largely Obsolete)**
  - An older system for typesetting Arabic-script languages, including Persian.
  - Difficult to use and outdated

An Example of Persian Xepersian

```latex
\documentclass{article}
\usepackage{graphicx}

\usepackage{xepersian}
\settextfont{B Nazanin}
\setlatintextfont{Times New Roman}
% \settextfont{B Zar}

\title{عنوان فارسی}

\begin{document}
    \maketitle
    \begin{abstract}
        این متن در بخش چکیده است.
    \end{abstract}

    \section{مقدمه}
    هدف از این درس آشنایی و کسب مهارت در اصول و روش‌های انجام تحقیق، اصول تهیه انواع ارائه‌های نوشتاری، مسائل مربوط به انواع ارائه‌های شفاهی و آشنایی با ابزارهای مربوطه می‌باشد.

    \section{روشها}
    \subsection{روش پیشنهادی اول}
    متن روش پیشنهادی اول.
    \subsection{روش پیشنهادی دوم}
    متن مربوط به روش پیشنهادی دوم با نام انگلیسی \lr{ABC}.

        \begin{latin}
        \section{Abstract}
        This is a paragraph written in English to show how xepersian handles
        multi-language documents. Notice how the alignment shifts to the left.
    \end{latin}

      دوباره به فارسی برگشتیم. شماره‌گذاری‌ها هم خودکار فارسی می‌شوند: ۱، ۲، ۳.
\end{document}
```


عنوان فارسی

۲۸ اردیبهشت ۱۴۰۵

چکیده

این متن در بخش چکیده است.

۱ مقدمه

هدف از این درس آشنایی و کسب مهارت در اصول و روش‌های انجام تحقیق، اصول تهیه انواع ارائه‌های نوشتاری، مسائل مربوط به انواع ارائه‌های شفاهی و آشنایی با ابزارهای مربوطه می‌باشد.

۲ روشها

۱.۲ روش پیشنهادی اول

متن روش پیشنهادی اول.

۲.۲ روش پیشنهادی دوم

متن مربوط به روش پیشنهادی دوم با نام انگلیسی ABC.

3 Abstract

This is a paragraph written in English to show how xepersian handles multi-language documents. Notice how the alignment shifts to the left.

دوباره به فارسی برگشتیم. شماره‌گذاری‌ها هم خودکار فارسی می‌شوند: ۱، ۲، ۳.

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Thank You